Output-Distribution Distillation for Contextual Sparsity Prediction in SwiGLU LLMs

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Abstract

Learned sparsity predictors for SwiGLU LLMs universally use binary cross-entropy (BCE), training each neuron independently, yet inference allocates budgets globally via top-k ranking. We train lightweight predictors (<1% of base parameters) with forward KL divergence, coupling all mask decisions through the output distribution. On LLaMA-3.1-8B at 30% sparsity, the KL predictor reduces perplexity by 14.5% over TEAL at matched sparsity, with commonsense gains but knowledge-intensive decline; post-hoc compensation (+67.2M params) becomes redundant. Output-distribution optimization yields masks that diverge 46–68% from magnitude-based targets; a target-swap ablation confirms these discovered patterns—not the training mechanism itself—as the proximate quality driver. A coupling alignment ranking emerges across five loss configurations; a per-layer KL control decomposes this into two dimensions with a task-dependent spectrum: signal type dominates PPL (98% of the gap at 30%), while coupling scope contributes 24–35% on downstream tasks—both dimensions carry independent value. Cross-architecture evaluation on Mistral-7B-v0.3 and Qwen-2.5-14B confirms PPL generalization with scale-dependent downstream effects.

1 Introduction

How the training objective shapes learned sparsity prediction quality is an open question for SwiGLU LLMs (Shazeer, 2020). SwiGLU FFN blocks exhibit high contextual sparsity—a property exploitable for potential inference acceleration—but their gated activation makes importance prediction challenging: neuron importance depends on both gate and up-projection values, a target that magnitude-based methods handle suboptimally. Learned sparsity predictors (Liu et al., 2023; Gautam et al., 2026) exploit this property by training small networks to anticipate which neurons are important, but they universally rely on binary cross-entropy (BCE) loss, treating each neuron as an independent classification target. Yet at inference, sparsity budgets are allocated globally: the TEAL (training-free activation sparsity; Liu et al., 2025) top-k mechanism ranks all neurons across all layers and retains only those above a model-wide threshold. This mismatch between per-neuron training and global inference allocation has not been formally analyzed. While MaskLLM (Fang et al., 2024) applies KL-based mask optimization, it targets static N:M weight sparsity; activation sparsity requires masks that generalize to unseen inputs, vary per token, and coordinate across layers through global allocation.

Gradient analysis reveals a structural asymmetry: BCE's gradient $\partial \mathcal{L}_{\text{BCE}}/\partial s_j = \sigma(s_j) - m_j^*$ depends only on neuron j, while under KL divergence, the gradient couples each neuron to all others through the downstream Jacobian (§3.6). KL's coupled gradients discover neuron importance patterns that diverge 68% from magnitude-based masks (per-layer Jaccard 32%). A target-swap ablation confirms that these discovered mask targets are the proximate quality driver at inference: BCE trained on KL-derived targets reproduces KL quality (max $\Delta \leq 0.79$pp at 50%, s42), while magnitude-trained BCE achieves 28.13 ± 0.20. Cross-layer gradient coupling is not the primary driver of PPL-level quality, but contributes meaningfully to downstream task fidelity—a PPL–downstream dissociation quantified in §4.5.

We instantiate a lightweight predictor (75.9M params, <1% of base) trained with forward KL and Gumbel-Sigmoid relaxation (Jang et al., 2017) on LLaMA-3.1-8B (Grattafiori et al., 2024). At 30% sparsity under uniform allocation, the predictor reduces TEAL's perplexity by 14.5% at matched sparsity (8.81 vs. 10.30 at 28.5%) with stable downstream (MMLU 40.7 ± 1.3%, 3-seed); global top-k

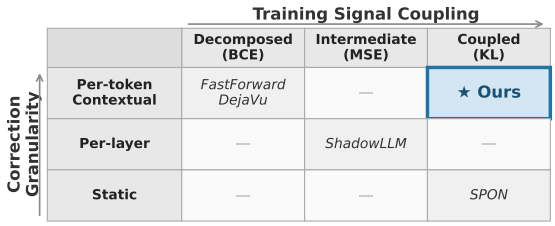


Figure 1: Method overview: a lightweight per-layer predictor produces importance scores trained via KL divergence with Gumbel-Sigmoid relaxation. At inference, scores feed TEAL's global top-k allocation without compensation networks.

amplifies cross-seed variance (7.3×). At 50%, the perplexity advantage widens (50% over TEAL), but downstream reasoning collapses regardless of method.

Cross-architecture evaluation on Mistral-7B-v0.3 (Jiang et al., 2023) (3-seed) confirms the PPL advantage (−17.3%) with robust commonsense gains but tied MMLU; preliminary evaluation on Qwen-2.5-14B (Team, 2024) shows consistent PPL improvement but reasoning decline (GSM8K −20pp, MMLU −2.8pp). This paper is primarily an empirical study: we do not propose a new architecture or algorithm, but rather characterize how the training objective shapes sparsity prediction quality, and analyze interactions between allocation strategy, sparsity level, and model scale. The empirical finding—that coupling alignment with the output distribution is the critical design dimension for learned sparsity—is the paper's central contribution. Wall-clock speedup requires sparse kernel support not addressed here (Appendix A).

Our contributions are:

- A coupling alignment ranking across five losses, decomposed by a per-layer KL control into signal type (98% of PPL gap) and coupling scope (24–35% on downstream tasks); output-distribution losses outperform intermediate losses regardless of coupling scope, with exploratory cross-scale evidence at 7–14B (§4).
- A target-swap ablation confirms that KL's discovered mask targets drive inference-stage quality; cross-layer coupling refines downstream fidelity beyond what PPL captures (§4.5).
- These mask patterns render post-hoc compensation (+67.2M params) redundant at both tested sparsity levels (§4.3).

2 Related Work

Training-Free Activation Sparsity. TEAL (Liu et al., 2025) applies global top-k selection by activation magnitude and achieves 40–50% sparsity with no training cost. CATS (Lee et al., 2024) adapts per-layer thresholds for SiLU/SwiGLU activations; R-Sparse (Zhang et al., 2025) predicts sparsity from rank-aware SVD projections; DIP (Federici et al., 2024) combines dynamic input pruning with cache-aware masking; SparseInfer (Shin et al., 2025) predicts activation sparsity patterns from attention outputs without additional training. Wang et al. (Wang et al., 2025) address dynamic sparsity–quantization interaction; AWQ (Lin et al., 2024) applies activation-aware weight quantization for on-device deployment. Dhar et al. (Dhar et al., 2024) characterize activation sparsity opportunities across architectures, showing that SwiGLU models exhibit lower natural sparsity than ReLU counterparts. OWL (Yin et al., 2024) determines non-uniform per-layer sparsity ratios based on activation outlier distribution for weight pruning. All training-free methods determine masks from fixed statistics or input-independent criteria, while our predictor generates *input-dependent* masks trained to minimize output-distribution divergence.

Predictor-Based Contextual Sparsity. DejaVu (Liu et al., 2023) introduced learned sparsity predictors trained via BCE to classify each neuron independently. ShadowLLM (Akhauri et al., 2024) uses attention-based routing with per-layer MSE supervision for intermediate coupling within each layer but without end-to-end gradient signal. FastForward (Gautam et al., 2026) adds a 67.2M-parameter compensation network using BCE for mask prediction, adding inference latency. SVD-CSP (Serbin et al., 2026) compresses sparsity predictors via SVD decomposition for reduced predictor overhead. Dhar et al. (Dhar et al., 2025) compress activation patterns via clustering for efficient neuron-level prediction. PowerInfer (Song et al., 2024a) and PowerInfer-2 (Xue et al., 2024) exploit power-law activation frequencies for GPU–CPU/NPU scheduling. MaskLLM (Fang et al., 2024) shares the principle of frozen weights + learnable masks + KL training, but the two methods differ in three fundamental ways. First, *sparsity type*: MaskLLM targets N:M semi-structured *weight* sparsity (static 2:4 patterns), while we target *activation* sparsity (dynamic per-token masks). Weight masks are fixed after

training; our masks vary with each input token. Second, *mask generation*: MaskLLM's masks are directly learnable parameters (one per weight group) selected via Gumbel-Softmax; we train a predictor network that generates masks at inference from hidden states for input-dependent decisions. Third, *allocation*: MaskLLM optimizes within fixed N:M groups where KL coupling operates only within small local neighborhoods; our masks are ranked under a global top-k budget across all layers, creating long-range dependencies that per-neuron BCE cannot capture and amplifying the coupled-vs-decomposed gap. The recipe-level principle (frozen weights + learnable masks + KL) overlaps with MaskLLM; our contribution lies in the analysis framework—target-swap ablation, signal type $\times$ coupling scope decomposition (§4.5)—applied to input-dependent activation sparsity under global allocation. All prior predictor-based methods for activation sparsity use decomposed (BCE) or intermediate (MSE) objectives where each neuron's gradient is independent of other mask states. Our contribution is primarily empirical: we observe that a coupled KL objective propagates gradients through all downstream layers and quantify when this matters for global allocation quality.

Static Bias Correction. SPON (Spontaneous Neurons; Xu et al., 2025) learns input-independent bias vectors via closed-form KL minimization per layer, absorbed into weights at deployment. Our method is fundamentally different: a per-token contextual predictor trained end-to-end across all layers that produces input-dependent masks. SPON solves a static per-layer problem; we solve a global dynamic prediction problem. The two are weakly correlated (cosine similarity 0.196, R^2=0.038; §4).

Activation Function Engineering. ReLU Strikes Back (Mirzadeh et al., 2024) replaces SiLU with ReLU for >90% natural sparsity; TurboSparse (Song et al., 2024b) and ProSparse (Song et al., 2025) engineer extreme sparsity through activation modifications; the Sparsing Law (Luo et al., 2024) formalizes how activation density increases with training compute. MiniTron (Muralidharan et al., 2024) combines structured pruning with distillation but modifies architecture permanently. Haziza et al. (Haziza et al., 2025) demonstrate 2:4 semi-structured activation sparsity with hardware-native acceleration, complementing our unstructured approach. MoC (Wu et al., 2025b) selectively activates Top-K FFN channels using SwiGLU's native gating during pretraining. All require retraining or weight modification; our method is a plug-in predictor on *unmodified* SwiGLU models.

Weight Pruning, Distillation, and Routing. Wanda (Sun et al., 2024), SparseGPT (Frantar and Alistarh, 2023), and SliceGPT (Ashkboos et al., 2024) produce static weight masks via one-shot or structured pruning; Polar Sparsity (Shrestha et al., 2025) scales contextual sparsity to batched inference by shifting focus from MLP to attention head sparsity. Switch Transformers (Fedus et al., 2022) and Mixture-of-Depths (Raposo et al., 2024) achieve conditional computation through learned routing. KL-based distillation (Kim and Rush, 2016; Sanh et al., 2019; Gu et al., 2024; Agarwal et al., 2024; Ko et al., 2024) trains compact students; SparseLoRA (Khaki et al., 2025) exploits contextual sparsity for fine-tuning. We apply the KL principle to train a sparsity predictor on unmodified dense models, using Gumbel-Sigmoid sampling (Maddison et al., 2017; Louizos et al., 2018) for differentiable discrete mask optimization.

MoE Routing and Gated Sparsity. Expert Choice routing (Zhou et al., 2022) lets each expert select its top-k tokens under a fixed capacity budget, a global allocation perspective that parallels TEAL's neuron-level global top-k. D2D-MoE (Szatkowski et al., 2024) converts dense SwiGLU layers into dynamic-k MoE by exploiting the same activation sparsity we target, but restructures the network into discrete experts rather than predicting masks on the original architecture. MoNE (Cheng et al., 2025) applies neuron-granular top-k selection within each MoE expert during pretraining for fine-grained conditional computation without additional routing parameters.

3 Method

We present a KL distillation framework for training contextual sparsity predictors in SwiGLU-based language models (Figure 1). The method replaces per-neuron binary classification (BCE) with an output-distribution objective that couples all pruning decisions through the model's forward pass and achieves competitive performance without auxiliary error compensation networks at inference.

3.1 Problem Formulation

Consider a pretrained transformer with L layers, each containing a SwiGLU feed-forward network (Shazeer, 2020). For layer ℓ, the FFN computes:

$$\mathbf{y}^{(\ell)} = \big(\sigma(\mathbf{x}^{(\ell)}\mathbf{W}_g^{(\ell)}) \odot \mathbf{x}^{(\ell)}\mathbf{W}_u^{(\ell)}\big)\mathbf{W}_d^{(\ell)}, \quad (1)$$

where $\mathbf{x}^{(\ell)} \in \mathbb{R}^d$ is the hidden state after attention (d=4096), σ denotes SiLU, $\mathbf{W}_g^{(\ell)}, \mathbf{W}_u^{(\ell)} \in \mathbb{R}^{d \times d_{\text{ff}}}$ and $\mathbf{W}_d^{(\ell)} \in \mathbb{R}^{d_{\text{ff}} \times d}$ are the gate, up-projection, and down-projection matrices, and d_{ff}=14,336.

Define the gate activation $\mathbf{g}^{(\ell)} = \sigma(\mathbf{x}^{(\ell)}\mathbf{W}_g^{(\ell)}) \in \mathbb{R}^{d_{\text{ff}}}$. Each element $g_j^{(\ell)}$ determines the contribution of neuron j to the output. We introduce a binary mask $\mathbf{m}^{(\ell)} \in \{0,1\}^{d_{\text{ff}}}$ that zeros out pruned neurons:

$$\mathbf{y}_{\text{sparse}}^{(\ell)} = \big(\mathbf{m}^{(\ell)} \odot \mathbf{g}^{(\ell)} \odot \mathbf{x}^{(\ell)}\mathbf{W}_u^{(\ell)}\big)\mathbf{W}_d^{(\ell)}. \quad (2)$$

Our goal is to learn a mask predictor $f_\theta^{(\ell)}(\mathbf{x}^{(\ell)}) \to \mathbf{s}^{(\ell)} \in (0,1)^{d_{\text{ff}}}$ that produces per-neuron importance scores from $\mathbf{x}^{(\ell)}$. At inference, the scores are fed to a global top-k allocation mechanism (Liu et al., 2025) that distributes the sparsity budget across layers non-uniformly, retaining the $(1-\rho) \times L \times d_{\text{ff}}$ highest-scoring neurons model-wide for a target sparsity ratio ρ.

3.2 Predictor Architecture

Each layer ℓ has an independent two-layer MLP predictor:

$$f_\theta^{(\ell)}(\mathbf{x}^{(\ell)}) = \text{GELU}(\mathbf{x}^{(\ell)}\mathbf{W}_1^{(\ell)})\,\mathbf{W}_2^{(\ell)}, \quad (3)$$

where $\mathbf{W}_1^{(\ell)} \in \mathbb{R}^{d \times d_b}$ and $\mathbf{W}_2^{(\ell)} \in \mathbb{R}^{d_b \times d_{\text{ff}}}$ with bottleneck dimension d_b=128. The predictor takes $\mathbf{x}^{(\ell)}$ (the hidden state after attention, before the FFN) and outputs d_{ff}-dimensional logits $\mathbf{s}^{(\ell)}$, from which the mask is derived.

Each predictor has $(4096 \times 128) + (128 \times 14336) = 2.37$M parameters. With 32 independent predictors (one per layer), the total overhead is 75.9M parameters, less than 1% of the 8B base model. The predictors operate independently across layers, requiring no cross-layer communication at inference.

3.3 Gumbel-Sigmoid Relaxation

The hard mask $m_j = \mathbf{1}[s_j > \text{threshold}]$ is non-differentiable, preventing gradient flow from the KL objective through the mask to the predictor parameters. We apply Gumbel-Sigmoid relaxation (Maddison et al., 2017; Louizos et al., 2018; Jang et al., 2017) to produce differentiable soft masks:

$$\tilde{m}_j = \sigma_\tau\left(\log \frac{s_j}{1-s_j} + g_1 - g_2\right), \quad g_1, g_2 \sim \text{Gumbel}(0,1), \quad (4)$$

where $\sigma_\tau(x) = \text{sigmoid}(x/\tau)$ and τ is a temperature parameter. The Gumbel noise provides stochastic exploration of mask configurations during training, while the temperature controls sharpness: as $\tau \to 0$, $\tilde{m}_j$ approaches a binary decision.

We anneal τ linearly from 1.0 to 0.1 over training. At high temperature, the soft mask permits gradient flow through all neurons for broad exploration. As temperature decreases, the mask concentrates on confident decisions, approximating the hard allocation used at inference.

Comparison with Straight-Through Estimation. The straight-through estimator (STE) (Bengio et al., 2013) applies a hard threshold in the forward pass and passes gradients through unchanged ($\partial m/\partial s = 1$). In our setting, STE produces discontinuous gradient signals: small parameter changes can flip many neurons simultaneously. The resulting oscillations in the penalty weight prevent convergence to the target sparsity. Gumbel relaxation eliminates this pathology and contributes a 33.8% improvement (PPL 15.77 → 11.79 at 50% sparsity), as we verify empirically in §4.

3.4 KL Divergence Training Objective

Let $\mathbf{z}_{\text{dense}}$ denote the output logits from a full dense forward pass, and $\mathbf{z}_{\text{sparse}}$ the logits from a forward pass with predictor-generated soft masks $\{\tilde{\mathbf{m}}^{(\ell)}\}_\ell$ applied at every layer. We minimize the forward KL divergence between the dense and sparse output distributions:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(p_{\text{dense}} \,\|\, p_{\text{sparse}}) = \sum_v p_{\text{dense}}^{(v)} \log \frac{p_{\text{dense}}^{(v)}}{p_{\text{sparse}}^{(v)}}, \quad (5)$$

where $p_{\text{dense}} = \text{softmax}(\mathbf{z}_{\text{dense}})$ and $p_{\text{sparse}} = \text{softmax}(\mathbf{z}_{\text{sparse}})$, summed over the vocabulary v and averaged over sequence positions. The base model is frozen; only the predictor parameters receive gradients.

We use forward KL (mass-covering) rather than reverse KL, as the goal is to preserve the

dense model's primary predictions rather than explore alternative modes. In sparsity prediction, mode-covering is critical: false negatives (omitting a rarely but critically active neuron) cascade through downstream layers under global allocation, whereas false positives merely waste capacity; reverse KL's mode-seeking concentrates on high-probability neurons and risks missing compositionally important activations (Wu et al., 2025a). §4 validates this choice empirically: forward KL outperforms reverse KL by 8.6% and JSD by 6.4% (Table 4). No temperature scaling (T=1) is applied (Hinton et al., 2015).

3.5 Adaptive Penalty Controller

We enforce a model-level sparsity target ρ via constrained optimization with $\bar{s} = \frac{1}{L}\sum_\ell \frac{1}{d_{\text{ff}}}\sum_j \tilde{m}_j^{(\ell)}$ denoting the average retained fraction:

$$\min_\theta \ \mathcal{L}_{\text{KL}} \quad \text{s.t.} \quad \bar{s} = 1 - \rho. \tag{6}$$

We relax this via an adaptive penalty controller:

$$\mathcal{L} = \mathcal{L}_{\text{KL}} + \lambda \cdot \text{clamp}(|\bar{s} - (1-\rho)| - \epsilon, \min{=}0)^2, \tag{7}$$

where $\epsilon = 0.02$ is a deadzone margin. The penalty weight λ is updated via geometric doubling (doubled when sparsity exceeds target, halved otherwise), clamped to $[\lambda_{\text{floor}}, \lambda_{\max}]$, reaching target sparsity within 40–110 steps across all tested regimes (Appendix C).

3.6 Gradient Analysis: BCE vs. KL

To build intuition for how KL distillation discovers neuron importance patterns that diverge from magnitude-based masks (§4.4), we contrast the gradient structure of the two objectives. This analysis is suggestive; §4.4 provides evidence that the coupled gradient discovers mask targets that yield lower perplexity, with mask quality driving inference-stage quality.

Under BCE, the predictor trains against oracle masks $m_j^* = \mathbf{1}[|g_j| \geq \text{top-}k \text{ threshold}]$ derived from activation magnitudes. The gradient $\partial\mathcal{L}_{\text{BCE}}/\partial s_j = \sigma(s_j) - m_j^*$ depends only on neuron j; each neuron is trained independently with no inter-neuron coupling. Under KL distillation, the chain rule introduces a downstream Jacobian $\partial \mathbf{z}_{\text{sparse}}/\partial \tilde{m}_j^{(\ell)}$ that couples neuron j's gradient to all other retained neurons in layers ℓ through L, because the sparse hidden state propagates through subsequent attention and FFN operations. This coupling allows KL to evaluate each neuron's importance in the context of the full sparse computation, rather than against a fixed magnitude target.

Table 1: Taxonomy of learned sparsity prediction methods along correction granularity (rows) and training signal coupling (columns). Our method occupies the Per-token × Coupled cell (organizational framework). Empty cells represent unexplored combinations; their relative merit is an open question.

	Decomposed (BCE)	Intermediate (MSE)	Coupled (KL)
Static	—	—	SPON
Per-token	DejaVu, FastForward	ShadowLLM, D2DMoE	**Ours**

Gradient SNR measurements[1] reveal that BCE maintains consistently higher SNR than KL throughout training (2.3× at convergence; Figure 5), indicating that KL's quality advantage does not arise from cleaner gradient signals. Rather, KL's end-to-end coupled gradient serves as a *search mechanism* that discovers compositionally superior mask patterns, even when the per-step signal is noisier than BCE's decomposed updates. The target-swap ablation in §4.4 confirms that the discovered mask targets, not the coupling mechanism itself, drive inference-stage quality.

3.7 Two-Dimensional Taxonomy of Sparsity Prediction Methods

We organize prior work along two axes: *correction granularity* (how the mask adapts to input) and *training signal coupling* (whether the objective captures inter-neuron dependencies).

Our method combines per-token prediction with coupled KL training (Table 1). KL on output logits directly optimizes next-token prediction loss, which composes across layers; per-layer MSE does not—a mask minimizing FFN output error at layer ℓ may amplify errors in layers $\ell{+}1 \ldots L$, explaining its 1.7×/9.6× worse PPL than TEAL at 30%/50% (§4). Full taxonomy discussion in Appendix O.

4 Experiments

4.1 Experimental Setup

Model and Predictor. We evaluate on LLaMA-3.1-8B (Grattafiori et al., 2024), which uses SwiGLU FFN blocks (Shazeer, 2020) with $d_{\text{model}} = 4096$ and $d_{\text{ff}} = 14336$ across 32 layers. The sparsity predictor is a two-layer MLP

[1] $\text{SNR}_\ell = \|\mathbb{E}[\nabla_{\theta_\ell}\mathcal{L}]\|_2^2 / \text{Var}[\nabla_{\theta_\ell}\mathcal{L}]$, expectation over mini-batches.

with 2.37M parameters per layer (75.9M total), implemented as Linear($4096 \rightarrow 128$) + GELU + Linear($128 \rightarrow 14336$), trained to produce per-neuron importance scores that minimize output-distribution divergence under sparsity.

Training. All predictors are trained for 1000 steps on WikiText-103 with sequence length 2048 and batch size 4 on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (96 GB), using AdamW (Loshchilov and Hutter, 2019) with learning rate 1×10^{-3}, weight decay 1×10^{-2}, and cosine decay. The KL predictor uses Gumbel-Sigmoid relaxation (Jang et al., 2017) with temperature $\tau = 1.0$ annealed to 0.1 over training.

Evaluation. We report perplexity on WikiText-2 (Merity et al., 2017) (seq_len=2048) and downstream accuracy on ARC-Challenge (25-shot), WinoGrande (5-shot), MMLU (5-shot), GSM8K (8-shot), HellaSwag (10-shot), and PIQA (0-shot).

Inference Protocol. At inference, the predictor produces per-neuron scores that are passed to TEAL's global top-k allocation mechanism (Liu et al., 2025), which distributes the sparsity budget across layers non-uniformly. We term this *predictor-TEAL*: the predictor replaces TEAL's magnitude-based scoring while retaining its allocation strategy.

Baselines. We compare against: (1) **Vanilla TEAL** (Liu et al., 2025), which uses activation magnitudes directly[2]; (2) **BCE predictor** (DejaVu-style (Liu et al., 2023)), trained with per-neuron binary cross-entropy on magnitude-based oracle masks for 1000 and 2700 steps; and (3) **Uniform allocation**, which distributes the sparsity budget equally across layers.

4.2 Main Results

Table 2 presents the primary comparison. Uniform allocation preserves downstream performance best; global allocation is analyzed to understand allocation-quality tradeoffs (§4.6). At 30% sparsity, KL distillation under uniform allocation achieves PPL 8.81±0.02 (3-seed), reducing TEAL's perplexity by 19% (10.82 → 8.81; 14.5% at matched 28.5% sparsity, Appendix D). Uniform allocation yields stable downstream (MMLU 40.7±1.3%, 3-seed; Appendix F), while global top-k exhibits high seed variance (35.7±9.5%) from late-layer starvation (L31 retention 28.4%; §4.6).

[2]TEAL is deterministic; single-run evaluation suffices. KL/BCE predictors require multi-seed evaluation due to random initialization.

Table 2: Main results on LLaMA-3.1-8B (WikiText-2 PPL↓; downstream accuracy↑). **Bold**: best per sparsity. *Extended training budget (2.71× KL step budget), no compensation; training details in §4.3. †BCE 30%: TEAL global (2700 steps). ±: 3-seed mean±std ($s \in \{42, 123, 456\}$); unmarked: seed 42. TEAL uses uniform allocation at all sparsity levels. MSE: per-layer FFN output reconstruction loss (ShadowLLM-style (Akhauri et al., 2024); same architecture/data/optimizer, $\lambda_{\max}$=1000; Appendix G), 3-seed PPL (σ=1.68 vs. KL σ=0.02); downstream: seed 42. ‡**Actual sparsity at 30%**: KL 28.5% vs. TEAL 29.3% (matched-sparsity comparison in Appendix D); both within 0.5pp at 50%.

Sparsity	Method	PPL↓	ARC-c	WG	MMLU	GSM8K
0%	Dense	6.12	58.28	78.06	65.27	48.75
30%	TEAL uniform	10.82	39.33	65.04	**45.86**	**8.79**
	BCE global†	11.25	41.2	66.0	36.3	6.0
	MSE uniform	18.52±1.68	31.23	60.38	31.79	1.97
	KL global (Ours)‡	8.96±0.02	**43.8±3.6**	**70.2±0.9**	35.7±9.5	8.0±0.7
	KL uniform‡	**8.81±0.02**	43.7±2.5	68.9±0.4	40.7±1.3	7.5±0.1
50%	TEAL uniform	23.90	27.47	54.38	**29.26**	1.74
	BCE (2700 steps)*	28.13±0.20	27.6±0.6	51.8±0.6	26.1±0.5	2.1±0.2
	MSE uniform	229.26	25.00	50.59	23.79	1.59
	KL global (Ours)	**11.84±0.15**	32.85	**63.46**	28.1	1.29
	KL uniform	11.92±0.07	**33.8±1.2**	62.9±0.1	28.0±0.6	**2.2±0.1**

Task-Type Split. KL outperforms TEAL on commonsense reasoning (ARC-c, WG, HellaSwag, PIQA: 4/4 wins at 30% uniform) but underperforms on knowledge-intensive tasks (MMLU, GSM8K: 0/2). Forward KL's mode-covering property ($\mathbb{E}_p[\log p/q]$) preserves diverse activation patterns that benefit broad commonsense inference; however, this may dilute task-specific knowledge pathways that magnitude-based methods preserve more faithfully (hypothesis; mechanistic investigation left to future work).

At 50% sparsity, the PPL advantage widens: 11.92 vs. 23.90 (50% reduction over TEAL). The predictor generalizes to C4 validation at both 30% (PPL 16.03±0.01, 2-seed, global allocation) and 50% (52.5% over TEAL; Appendix D). However, all methods exhibit downstream collapse on MMLU (28.1±0.2% vs. 28.0±0.6%, both ≈ random, 3-seed) and GSM8K (<2.2%); this is a capacity-limited regime where no allocation strategy can recover.

4.3 Extended Training Analysis

To test whether BCE simply requires more compute, we train it for 2700 steps (2.71× KL budget, matched wall-clock: ∼46 min vs. ∼45 min on RTX PRO 6000). BCE achieves PPL 28.13±0.20 (3-seed), still 2.39× worse than KL (11.79). The

Table 3: Target-swap ablation (C1): BCE on KL-derived masks vs. KL predictor (same seed), TEAL uniform. All seeds show max $\Delta \leq 1.70$pp at 30%; max $\Delta \leq 0.79$pp at 50% (s42). †s42 oracle is from an earlier training run; s123/s456 use matched retrained oracles.

Sparsity	Seed	Method	PPL↓	ARC-c	WG	MMLU	GSM8K
30%	s42†	C1	9.45	40.27	67.40	**42.30**	**7.58**
		KL	**8.80**	**40.78**	**68.59**	40.6	7.35
	s123	C1	8.84	44.71	**69.77**	**39.92**	6.82
		KL	**8.83**	**45.31**	69.38	39.4	**7.51**
	s456	C1	8.82	44.71	**69.93**	41.04	6.60
		KL	**8.79**	**45.05**	68.75	**42.0**	**7.58**
50%	s42	C1	11.96	32.17	**63.14**	27.79	**2.05**
		KL	11.96	**32.59**	62.35	**27.92**	1.97

difference is systematic: BCE achieves +7pp oracle F1 (0.613 vs. 0.540) with 139% worse PPL, and KL's near-random precision (0.51) indicates a mask optimized for distributional fidelity rather than magnitude agreement.

4.4 Mask Pattern Analysis

To understand *what* the KL predictor learns, we compare KL-derived masks against magnitude-based oracle masks (the BCE training targets) at 50% sparsity on WikiText-2. Per-layer Jaccard similarity averages 32% at 50% sparsity (64 samples, 131K tokens; per-token computation): the KL predictor yields neuron importance patterns that differ from activation magnitude (Jaccard distance 46% at 30%, 68% at 50%; Figure 7). Cross-seed KL Jaccard distance (36.3%; Appendix F.1) provides a noise baseline; at 30%, per-layer mean divergence (42.8%) shows ∼6.5 pp systematic component above inter-seed variation. At layer 16, 41% of neurons fall in disagreement quadrants where KL and magnitude rankings diverge (Appendix M, Figures 7–6). Despite this divergence, KL achieves PPL 11.79 vs. BCE's 28.13±0.20: output-distribution fidelity, not magnitude agreement, produces lower-perplexity masks under global allocation.

Per-layer divergence follows a U-shaped pattern: middle layers show lowest divergence (∼0.61 at 50%), with late layers peaking at 0.70–0.75.

Target-Swap Ablation: Target Quality vs. Gradient Coupling. To disentangle whether KL's advantage stems from its coupled gradient pathway or from the mask patterns it discovers, we train a standard BCE predictor on KL-derived oracle masks instead of magnitude-based masks (C1 target-swap ablation). Table 3 reports results under TEAL uniform allocation at matched sparsity.

At 50%, C1 matches KL within 0.79pp, while magnitude-trained BCE yields 2.4× worse PPL (Table 2). At 30%, retrained seeds show ≤1.18pp agreement; the s42 oracle from an earlier checkpoint has a 0.65pp gap but confirms the core finding. C1 demonstrates that KL-discovered mask patterns *transfer* their quality advantage to a simpler training mechanism (PPL within 0.79pp; downstream variation up to 4.5pp MMLU), confirming masks as the proximate quality driver. However, KL's coupled optimization remains necessary for *discovering* these patterns: a per-layer KL control (§4.5) shows that cross-layer coupling contributes marginally to PPL but substantially to downstream fidelity.

Table 4: Ablation on LLaMA-3.1-8B (WikiText-2 PPL↓, unified benchmark protocol; uniform allocation at 30%, global allocation at 50%). 30%: 3-seed mean±std ($s \in \{42, 123, 456\}$) where available; 50%: seed 42. ‡STE achieved only 20.8% model-level sparsity at 30% target due to tied sigmoid scores in the threshold-based mask; excluded from 30% comparison (see §4.5). §MSE 50%: s42 only; s123 catastrophic (PPL=15,789, 5.1% sparsity; Appendix G).

Configuration	30% PPL↓	50% PPL↓	Δ_{50}
KL-based variants			
Forward KL + Gumbel (Ours)	**8.81±0.02**	**11.79**	—
JSD (symmetric) + Gumbel	8.90±0.02	12.54	+0.75
Reverse KL + Gumbel	9.26±0.07	12.80	+1.01
Forward KL + STE	—‡	15.77	+3.98
Per-layer training			
Per-layer MSE	18.52±1.68	229.26§	+217.47
Per-layer KL + Gumbel	9.05±0.02	34.85	+23.06
BCE-based variants			
BCE (no compensation)	15.71	27.97	+16.18
BCE + Gumbel (no penalty)	—	32.22	+20.43
BCE + Gumbel + Penalty	—	38.02	+26.23
BCE + STE	—	60.54	+48.75

Soft-vs-Hard Confound. C1 also changes mask representation (hard vs. soft), but BCE+Gumbel (32.22) vs. standard BCE (27.97) confirms softness alone does not explain KL's advantage; trajectory-dependent confounds are discussed in Appendix P.

4.5 Ablation Studies

Table 4 isolates the contribution of each design choice at 30% and 50% sparsity.

Relaxation, Divergence, and SPON. At 50%, Gumbel-Sigmoid outperforms STE (Bengio et al., 2013) by 33.8% (11.79 vs. 15.77); forward KL outperforms JSD (+6.4%) and reverse KL (+8.6%), consistent with mode-covering (Gu et al., 2024). At 30%, forward KL (8.81) and JSD (8.90) are near-identical, both outperforming reverse KL (9.26) and dramatically better than MSE (18.52±1.68;

σ=1.68 vs. KL's σ=0.02, reflecting per-layer signal instability). BCE without compensation (15.71) confirms decomposed losses underperform at both regimes. The near-equivalence of KL and JSD suggests the specific divergence measure matters less than whether the loss optimizes through the output distribution; within output-distribution losses, cross-layer scope provides downstream benefits not predicted by PPL alone (§4.5). Reverse KL at 30% (3-seed) shows the same asymmetry: commonsense exceeds TEAL (ARC-c 43.7%, WG 69.0%) while MMLU falls below (39.2% vs. 40.7% KL, 45.9% TEAL).

Per-Layer KL: Decomposing Signal Type and Coupling Scope. To isolate cross-layer coupling, we train per-layer KL predictors that mask one random layer per step and compute KL on the full model output, preserving the output-distribution signal without cross-layer gradients. Per-layer KL achieves 9.05±0.02, within 2.7% of end-to-end KL (8.81±0.02), while per-layer MSE degrades to 18.52±1.68 (2.05× worse). For PPL, signal type[3] accounts for 98% of the gap. However, downstream decomposition[4] reveals a task-dependent spectrum: coupling scope contributes 30–35% on commonsense (ARC-c 30.5%, WG 35.2%), 24% on MMLU, 31% on GSM8K, and 67% on BoolQ (per-layer KL BoolQ variance is high; interpret cautiously). This explains why per-layer KL matches E2E PPL but underperforms downstream. At 50%, coupling scope's importance increases sharply: per-layer KL degrades to 34.85 vs. 11.92 for E2E KL (+192%), with coupling's share rising from 2.5% to 10.6% (4×; Table 4) and downstream falling to TEAL level—coupling transitions from optional refinement to essential coordination as sparsity increases. A confound: training masks one layer per step while inference applies all 32; the downstream gap may partly reflect this train–inference shift.

STE Failure and Loss–Relaxation Interaction. At 30%, STE yields only 20.8% effective sparsity (vs. target 30%), confirming Gumbel-Sigmoid is necessary. BCE+Gumbel (32.22) underperforms standard BCE (27.97); the loss effect (ΔPPL=20.4) is 5.1× the relaxation effect (ΔPPL=3.98).

[3]We use "signal type" as shorthand for the training objective's optimization target (output distribution vs. intermediate activations); this dimension also entails differences in gradient flow topology and optimization landscape that are not independently controlled.

[4]E2E KL and per-layer KL are 3-seed means; MSE downstream is seed 42 only (Table 2).

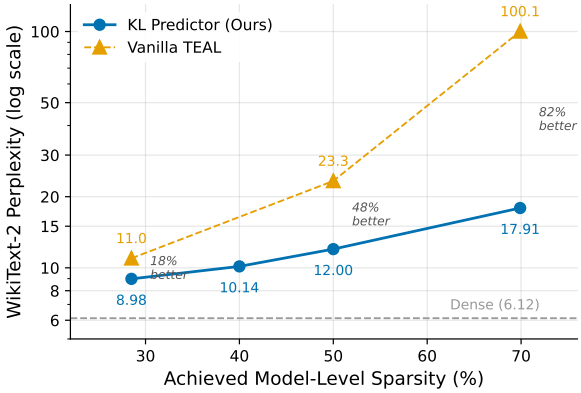


Figure 2: PPL scaling with sparsity (global allocation). KL's advantage grows: 14.5% at 30%, 50% at 50%, 82% at 70%.

Compensation Redundancy. FastForward-style compensation (Gautam et al., 2026) (+67.2M params) provides no benefit; robustness to d_b, temperature, and training steps confirmed in Appendices J–B.2.

4.6 Sparsity Scaling and Regime Analysis

Allocation Behavior. Table 14 (Appendix L) reveals non-monotonic allocation preference: uniform is best at ≤30% (7.3× smaller MMLU σ), global at 35–40%, both collapse at ≥45%.

Cross-Architecture and Scale. On Mistral-7B-v0.3 (Jiang et al., 2023), PPL generalizes (−17.3%, 3-seed; Table 17); on Qwen-2.5-14B (Team, 2024), PPL generalizes but MMLU/GSM8K decline (Table 13). KL's mode-covering property may preserve distribution breadth at the expense of reasoning pathways; the consistent regression across scales (8B MMLU −5.2pp, 14B GSM8K −20pp) suggests a systematic property of distribution-matching objectives. The pipeline adds 6.4% latency overhead (Appendix A).

5 Conclusion

Coupling alignment—optimizing through the output distribution—is the dominant design dimension for learned sparsity: signal type dominates PPL (98% of the gap at 30%), while coupling scope contributes 24–35% on downstream tasks—a task-dependent spectrum where both dimensions carry independent value. A predictor (<1% of base) reduces TEAL's perplexity by 14.5%, with KL-discovered masks (46–68% divergent from magnitude) driving inference quality; however, downstream transfer is task- and scale-dependent. Future work: N:M sparsity and 70B+ scaling.

Limitations

Our findings are bounded by three key limitations. **Sparsity sweet spot.** 30% appears to be the lowest-degradation level for downstream fidelity; at 40%, MMLU drops to 35.5% (vs. 40.7% at 30%), and at 50%, all methods collapse to near-random on reasoning tasks. A separate predictor must be retrained per target. **Knowledge-intensive regression.** KL consistently underperforms magnitude-based methods on MMLU and GSM8K across scales (8B: -5.2pp MMLU; 14B: -20pp GSM8K), suggesting distribution-matching objectives systematically sacrifice reasoning-critical pathways. **Training data homogeneity.** All predictors are trained on WikiText-103 (English Wikipedia). OOD evaluation on C4 shows preserved KL advantage but $31\times$ higher cross-seed variance (CV=6.3% vs. 0.2% in-domain), indicating domain sensitivity. Evaluation is limited to 7–14B; scaling to 70B+ and wall-clock speedup via sparse kernels remain future work.

Ethics Statement

This work proposes a training method for sparsity predictors in pretrained language models, evaluated on publicly available models (LLaMA-3.1-8B, Mistral-7B-v0.3, Qwen-2.5-14B (Team, 2024)) and standard benchmarks (WikiText-2, WikiText-103, C4, ARC, WinoGrande, HellaSwag, PIQA, MMLU, GSM8K). No new data collection, annotation, or human subjects are involved. The method does not alter model outputs beyond the sparsity-induced degradation we analyze; it introduces no new generation capabilities or safety risks beyond those inherent in the base models. Compute requirements are modest (1000 training steps, $\sim$45 minutes on a single GPU for KL training), presenting minimal environmental concern relative to the pretrained models we evaluate on. We release no model weights or user-facing systems; our contribution is analytical (understanding KL distillation for sparsity) rather than deployment-oriented.

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A Inference Efficiency

Table 5 reports inference throughput on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (96 GB; batch size 1, sequence length 2048, fp16, averaged over 100 forward passes).

Table 5: Inference latency on LLaMA-3.1-8B (RTX PRO 6000 Blackwell, batch=1, seq=2048, fp16, 100 forward passes). Overhead is relative to dense inference.

Method	Sparsity	ms/fwd	tok/s	Overhead
Dense	0%	119.11±0.57	17,195	—
KL predictor-only	30%	126.64±0.70	16,172	+6.3%
KL predictor-only	50%	126.78±0.31	16,154	+6.4%
Predictor + compensation	30%	129.94±0.75	15,761	+9.1%
Predictor + compensation	50%	129.71±0.53	15,789	+8.9%

The full KL-sparse pipeline adds 6.4% overhead to dense inference (126.78 vs. 119.11 ms/forward), of which the predictor's 32-layer forward pass contributes 3.17 ms (2.7%); the remainder arises from mask application and sparse FFN indexing. Eliminating the compensation network saves an additional ∼2.7% latency overhead and 67.2M parameters (the per-layer GEMV required for output rescaling); KL distillation thus provides both quality and efficiency advantages over BCE-based approaches. The overhead is near-identical at 30% and 50% sparsity because the predictor runs the same computation regardless of target sparsity; only the top-k threshold changes.

Note that the current implementation uses unstructured element-wise masks, which do not translate to wall-clock speedup without sparse kernel support. The pipeline adds ∼6.4% overhead; net acceleration requires kernel-level sparse computation optimizations (e.g., Triton block-sparse, cuSPARSE) that are beyond the scope of this work. Our contribution is prediction quality and inference simplification (removing the compensation network), not end-to-end latency reduction.

B Predictor Bottleneck Sensitivity

Table 6 reports the effect of predictor bottleneck dimension d_b on PPL at 50% sparsity under both TEAL global allocation (the paper's primary inference protocol) and per-token evaluation.

Table 6: Predictor bottleneck sensitivity at 50% sparsity (WikiText-2 PPL↓). TEAL global is the primary inference protocol; per-token shown for reference.

d_b	Params	TEAL global	Δ	Per-token
64	38.2M (0.48%)	12.076	+0.9%	10.16
128	75.9M (0.95%)	11.963	—	9.218
256	151.5M (1.89%)	12.105	+1.2%	8.586
512	302.5M (3.77%)	12.315	+2.9%	7.93

Under TEAL global allocation, increasing d_b beyond 128 provides no benefit (PPL slightly worsens), while per-token evaluation shows monotonic improvement (10.16→7.93). This divergence indicates that the global top-k allocation mechanism, not predictor capacity, is the binding constraint: additional predictor capacity produces better per-neuron scores, but the global ranking and budget distribution cannot exploit them. The d_b=128 configuration (75.9M params, <1% of base model) is therefore optimal for the TEAL inference protocol.

B.1 Temperature Schedule Sensitivity

Table 7: Temperature schedule sensitivity at 50% sparsity (WikiText-2 PPL↓, TEAL global).

τ Schedule	PPL↓
Fixed τ=0.5	**11.861**
Linear 1.0 → 0.1 (default)	11.963
Fixed τ=0.1	12.072

Temperature schedule has minimal impact (<1.8% PPL variation). The linear annealing default is a conservative choice; even removing annealing entirely (fixed τ=0.5) yields comparable performance.

B.2 Training Convergence

Table 8: Training convergence at 50% sparsity (WikiText-2 PPL↓, TEAL global).

Steps	PPL↓
200	15.158
400	12.538
600	12.046
800	11.907
1000	**11.906**

Training converges rapidly: 95% of the quality gap closes by step 600, and performance saturates by step 800 ($\Delta < 0.001$ between 800 and 1000 steps). Our default of 1000 steps provides a conservative margin.

C Adaptive Penalty Controller Dynamics

Table 9 reports the convergence of the adaptive penalty controller across sparsity targets. The geometric doubling pattern ($1 \to 2 \to 4 \to 8 \to \ldots$) enables λ to reach steady state within ∼10 steps; the deadzone ($\epsilon = 0.02$) then stabilizes λ once actual sparsity is within ±2% of the target.

Table 9: Penalty controller convergence across sparsity targets (multiplicative schedule, margin $\epsilon = 0.02$, 1000 training steps). Deviation is |actual − target| / target.

Target ρ	$\lambda_{\max}$	Final λ	Actual sparsity	Deviation
30%	5000	5000 (cap)	0.285	5.0%
40%	5000	625	0.398	0.5%
50%	1000	8	0.504	0.8%
70%	1000	1000 (cap)	0.688	1.7%

At 50% sparsity, the SwiGLU gate distribution already clusters near the target, so λ remains low (8); the constraint is nearly satisfied by the initial predictor outputs. At 30% and 70%, the gap between the gate prior and target is larger, driving λ

to its respective caps. The 40% target converges to $\lambda = 625$ with only 0.5% deviation; this precision reflects a regime where the target is reachable without λ saturation.

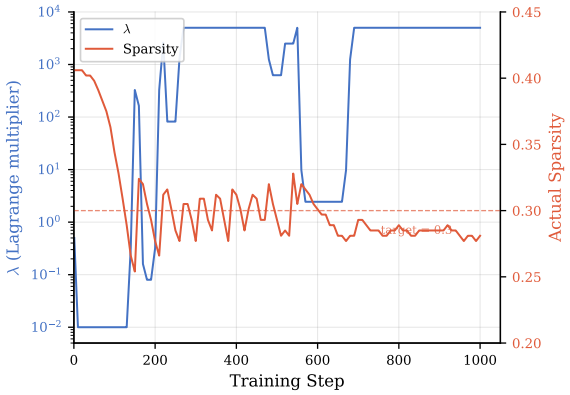


Figure 3: Lagrangian controller dynamics for 30% sparsity target ($\lambda_{\max} = 5000$). Left axis (log scale): penalty weight λ rises from 0.01 to its cap once actual sparsity undershoots the target at ∼step 150, exhibiting bang-bang oscillations before saturating. Right axis: actual sparsity (solid) converges from the natural density (≈0.41) toward the target (dashed) and stabilizes near 0.28.

Temperature τ is annealed linearly from 1.0 to 0.1 over 1000 steps. A mild loss increase in the final ∼100 steps is expected: as masks become more binary ($\tau \to 0$), discrete rounding noise slightly increases KL divergence, consistent with convergence toward the hard-mask inference regime.

The controller reliably drives actual sparsity to within 5% of the target across all tested regimes, with λ spanning three orders of magnitude (8 to 5000) via geometric doubling. For the 30% target, actual sparsity is ≈28.5% (5% deviation). This does not affect conclusions: TEAL at 28.5% actual sparsity yields PPL 10.30 (vs. 10.97 at exact 30%), a 0.67 PPL difference negligible relative to the KL vs. TEAL gap (8.96 vs. 10.97).

D Out-of-Distribution Evaluation

To address concerns about data homogeneity between training (WikiText-103) and evaluation (WikiText-2), we evaluate on C4 (Raffel et al., 2020) validation set under TEAL global allocation.

At the 30% sparsity level, the KL predictor achieves C4 PPL 16.03±0.01 (2-seed, global allocation, 1.74× dense), confirming OOD generalization. At 50%, the KL predictor achieves 52.5% relative advantage over TEAL (18.40 vs. 38.75), comparable to the 50% advantage on WikiText-2 (11.92 vs. 23.90), confirming that KL-trained predictions generalize across distributions.

Table 10: Out-of-distribution evaluation on C4 validation (PPL↓). 30% KL reports 2-seed mean±std; other entries use seed 42.

Sparsity	Method	C4 PPL↓
0%	Dense	9.21
30%	KL predictor (Ours)	**16.03±0.01**
50%	KL predictor (Ours)	**18.40**
	TEAL vanilla	38.75

Uniform Allocation. Under uniform allocation at 30%, C4 PPL across three seeds is 18.22 (s42), 16.37 (s123), 16.36 (s456), yielding mean 16.98±1.07 (1.84× dense). The cross-seed variance is substantially higher than in-domain: CV=6.3% on C4 vs. 0.2% on WikiText-2 (PPL 8.81±0.02), driven by seed 42's elevated C4 PPL (18.22 vs. 16.37 for other seeds). Global allocation's cross-layer budget flexibility partially compensates for domain shift (PPL 16.03 global vs. 16.98 uniform), consistent with the Limitations observation that the predictor partially overfits to training-domain activation patterns.

Downstream Task Robustness. Table 11 compares downstream performance when the predictor generates masks from C4 text (OOD) vs. WikiText (in-domain), both at 30% uniform allocation (seed 42). All six benchmarks show $\Delta \leq 1.1$pp, confirming that predictor masks are robust to calibration domain shift.

Table 11: Downstream task robustness to calibration domain shift (KL 30% uniform, seed 42). WikiText: predictor masks from in-domain text; C4: masks from OOD text. Metrics: acc_norm (ARC-c, HS, PIQA), acc (WG, MMLU), exact_match (GSM8K).

Calibration	ARC-c	WG	MMLU	GSM8K	HS	PIQA
WikiText	40.8	68.6	40.6	7.4	70.9	74.8
C4 (OOD)	40.4	69.3	41.4	6.3	70.7	74.0
Δ	−0.4	+0.7	+0.8	−1.1	−0.2	−0.8

E Per-Layer Retention Analysis

F Training Stability

Three-seed evaluation ($s \in \{42, 123, 456\}$) at 50% sparsity under uniform allocation yields PPL 11.92 ± 0.07 (CV=0.6%); training is stable across random initializations. Downstream metrics are similarly stable: ARC-c 33.8 ± 1.2%, WinoGrande

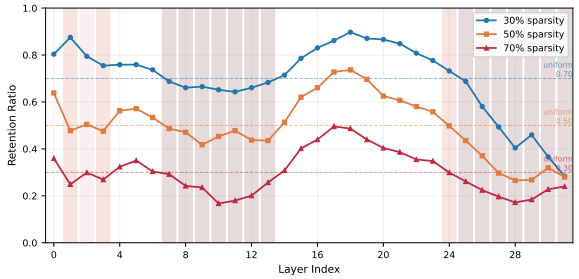


Figure 4: Per-layer neuron retention under global top-k allocation at 30%, 50%, and 70% sparsity. Dashed lines indicate uniform retention baselines. Late layers (L24–31) are consistently starved due to long-tailed activation distributions, with retention dropping to 28.4% (L31) even at 30% sparsity. At 70%, starvation extends to mid-layers (L10–12: 16.7–20.1%), explaining the transition to the capacity-limited regime.

$62.9 \pm 0.1\%$, MMLU $28.0 \pm 0.6\%$, GSM8K $2.2 \pm 0.1\%$.

Under global allocation at 50%, MMLU is equally stable: $28.1 \pm 0.2\%$ (s42=27.89, s123=28.34, s456=27.94); the near-random performance is therefore capacity-limited rather than seed-sensitive (σ=0.2pp vs. 9.5pp at 30% global). The low standard deviation across both allocation strategies indicates that the KL training landscape is well-conditioned for this predictor architecture, with low sensitivity to initialization.

F.1 Cross-Seed Mask Consistency

To assess whether KL predictors discover consistent mask patterns across initializations, we compute pairwise Jaccard distances among three independently trained KL predictors (s42/s123/s456) at 30% uniform allocation (100 sequences, WikiText-2 validation).

The mean cross-seed Jaccard distance is 0.363 (IoU = 0.637), substantially lower than the KL-vs-magnitude divergence (0.46–0.68), confirming that KL-discovered patterns form a distinct cluster from magnitude-based masks. Per-layer divergence follows a U-shaped pattern: lowest at embedding-adjacent layers (L0: 0.33), peaking at early layers (L1: 0.41), with a mid-network valley (L12–14: 0.33). The random baseline Jaccard distance is 0.462.

While specific mask configurations vary moderately across seeds (Jaccard distance 36.3%), the resulting PPL is highly stable (8.81 ± 0.02, CV=0.2%), suggesting multiple near-equivalent optima in the activation mask space. The cross-seed divergence (36.3%) is substantially lower than the KL-vs-magnitude divergence (46–68%), confirming that KL discovers effective patterns that cluster distinctly from magnitude-based selection.

G MSE Baseline Configuration

The MSE predictor uses identical architecture (two-layer MLP, bottleneck d_b=128), training data (WikiText-103, 1000 steps), optimizer (AdamW, lr=1×10^{-3}, weight decay 1×10^{-2}, cosine decay), Gumbel-Sigmoid mask generation (τ: $1.0 \to 0.1$), and adaptive Lagrangian sparsity control as the KL and BCE predictors. The only difference is the loss target: per-layer $\|\mathbf{W}_d^{(\ell)}(\tilde{\mathbf{m}}^{(\ell)} \odot \mathbf{g}^{(\ell)} \odot \mathbf{x}^{(\ell)}\mathbf{W}_u^{(\ell)}) - \mathbf{y}_{\text{dense}}^{(\ell)}\|_2^2$, replacing the end-to-end KL divergence. This controlled comparison isolates the effect of coupling alignment: all three predictors see identical data in identical order with identical architectures; only the loss function differs.

MSE uses $\lambda_{\max}$=1000 (vs. KL's 5000) because MSE's per-layer gradient magnitude is approximately $5\times$ larger than KL's end-to-end gradient, requiring proportionally less penalty pressure to achieve comparable sparsity control. Both reach their target sparsity within 110 steps. At 50% target, MSE converges to the target sparsity for seed 42; however, with seed 123, sparsity reaches only 5.1% under the same $\lambda_{\max}$ cap (eval PPL=15,789, random-level), indicating high sensitivity to initialization—a failure mode not observed with KL-based losses. At 30%, MSE achieves 27.7% actual sparsity for s42 (gap 2.3pp). The 3-seed MSE mean (18.52 ± 1.68; s42=20.43, s123=17.73, s456=17.39) is still $1.7\times$ worse than TEAL (10.82), confirming that the performance deficit stems from the loss function, not insufficient sparsity pressure.

H Extended SNR Analysis

Figure 5 plots gradient SNR across the full 1000-step training for both KL and BCE predictors, measured at 14 checkpoints (LLaMA-3.1-8B, C4) with 95% bootstrap confidence intervals. From step 100 onward, BCE maintains consistently higher SNR than KL (2.3–$17\times$ depending on training phase), with the gap narrowing toward convergence ($2.3\times$ at step 1000: 2.026×10^{-3} vs. 8.78×10^{-4}). Despite KL's noisier gradient signal, KL-trained predictors produce substantially better masks (Table 2), confirming that *target alignment* matters more than gradient clarity for mask quality. BCE's higher SNR reflects its simpler per-neuron classifi-

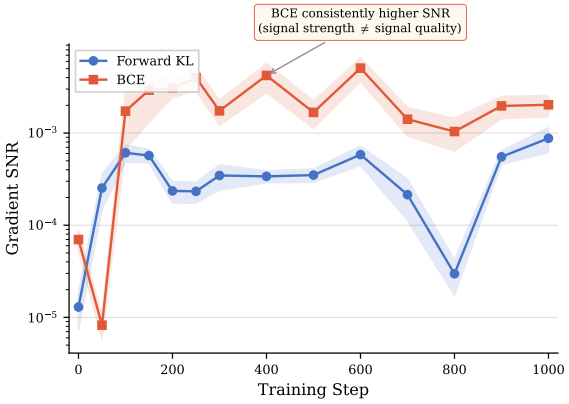


Figure 5: Gradient SNR trajectory across 1000 training steps. BCE (orange) maintains consistently higher SNR than KL (blue) from step 100 onward (2.3–17×), yet KL produces superior masks (Table 2), suggesting that target alignment matters more than gradient clarity for mask quality.

cation objective, which concentrates gradients on individual neuron decisions; KL's coupled gradient distributes signal across all mask decisions jointly, yielding lower SNR but discovering compositionally superior mask patterns (§3.6).

I Extended Downstream Evaluation

Table 12 reports HellaSwag and PIQA across sparsity levels and allocation strategies.

Table 12: Extended downstream benchmarks (LLaMA-3.1-8B). HellaSwag: 10-shot normalized acc; PIQA: 0-shot normalized acc. KL 30% uniform: 3-seed mean±std; others: seed 42.

Sparsity	Method	HellaSwag	PIQA
0%	Dense	**82.35**	**80.90**
30%	KL uniform	**71.07±0.54**	**74.95±0.25**
	TEAL uniform	63.31	72.80
	BCE isocompute	62.81	74.21
50%	KL global	**58.95**	**69.48**
	KL uniform	56.66	69.10
	TEAL uniform	39.35	63.66

At 30%, the KL predictor retains 86.3% of dense HellaSwag accuracy (71.07±0.54 vs. 82.35, 3-seed) and 92.6% of PIQA (74.95±0.25 vs. 80.90), confirming moderate degradation on non-reasoning tasks. At 50%, global allocation outperforms uniform on HellaSwag (+2.29pp), consistent with the budget-concentration benefit observed for PPL in the capacity-limited regime (§4.6); PIQA shows a smaller gap (+0.38pp).

J Compensation Training Convergence

All compensation networks (Gautam et al., 2026) were trained for 1000 steps (lr=10^{-4}, batch size 4, seq length 2048). Training loss plateaus by step 700 across all valid runs: 30% s42 (final loss 0.70), 30% s456 (0.68), 50% s42 (1.07), with fluctuations remaining within 0.08 loss units for the final 300 steps. The compensation network converges but provides no quality improvement over the KL predictor alone, confirming that the redundancy result reflects predictor self-sufficiency rather than insufficient compensation training.

K Qwen-2.5-14B Detailed Results

Table 13: Qwen-2.5-14B at 30% nominal sparsity. Actual sparsity differs: KL 29.6% (3-seed), TEAL 29.3% (target 30%). **Bold**: best sparse per column per group. BCE collapses to 6.4% native sparsity (§3.6), so forced global top-k is used.

Method	PPL↓	ARC-c	WG	MMLU	GSM8K	HS	PIQA
Dense	5.19	67.58	81.06	79.77	87.57	84.20	81.94
Global top-k allocation							
KL (Ours)	**7.62**	57.17	72.06	63.72	48.82	**75.01**	**77.37**
TEAL	8.73	56.83	**72.69**	**71.83**	49.51	70.51	74.70
BCE	8.76	**57.59**	72.61	71.67	**53.07**	53.43	68.28
Uniform allocation							
KL (Ours)	**7.90**	53.50	**72.82**	63.55	32.1±5.2	**71.67±1.0**	**75.06±0.2**
TEAL	8.01	**55.29**	69.85	**66.34**	**52.1**	67.72	74.70

Under global allocation, BCE's HellaSwag drops to 53.43% (−17pp vs. TEAL), indicating that the magnitude-based predictor's mask quality degrades severely when combined with non-uniform budget distribution at 14B scale.

L Full Allocation Analysis

Table 15 extends Table 14 with all downstream metrics.

M Mask Pattern Visualization

Figure 7 shows per-layer Jaccard distance between KL-derived and magnitude-based masks at 30% and 50% sparsity. Figure 6 shows per-neuron importance scores at layer 16. Figure 8 shows binary mask comparisons at layers 2, 16, and 28.

N Cross-Architecture Comparison

Table 16 evaluates on Mistral-7B-v0.3 (Jiang et al., 2023) (GQA + sliding window, dense PPL 5.52).

The KL PPL advantage ratio is consistent across architectures (within 8% at ≤50% sparsity). Table 17 reports downstream task accuracy at 30% sparsity on Mistral-7B-v0.3 (3-seed).

Table 14: Allocation summary (LLaMA-3.1-8B). ±: 3-seed; *high seed variance.

Sparsity	Allocation	PPL↓	MMLU
30%	Global	8.96±0.02	35.7±9.5*
	Uniform	**8.81±0.02**	**40.7±1.3**
50%	Global	**11.79**	**28.1±0.2**
	Uniform	11.92±0.07	28.0±0.6

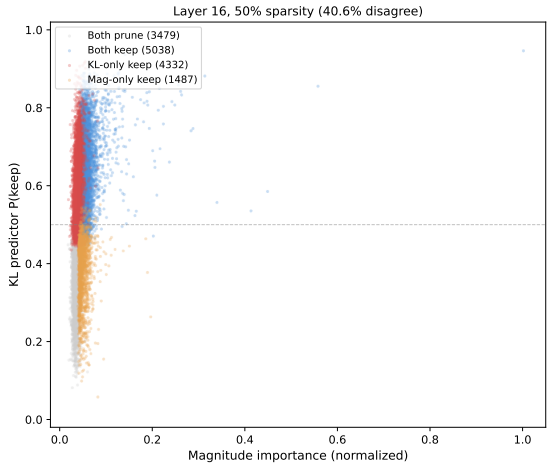


Figure 6: KL vs. magnitude importance scores for layer 16 neurons at 50% sparsity. 41% of neurons fall in disagreement quadrants where one method selects a neuron the other discards.

KL improves PPL (−17.3%) and commonsense tasks (WG +6.7pp) but MMLU is tied (43.3% vs. 43.5%); ARC-c shows high seed variance (σ=6.4pp); GSM8K reverses.

O Taxonomy Discussion

Correction granularity. *Static* methods (Xu et al., 2025) learn a fixed bias vector per layer that does not depend on the input token. *Per-token* methods (Liu et al., 2023; Gautam et al., 2026; Akhauri et al., 2024; Szatkowski et al., 2024) produce a unique mask for each input position, conditioning on the token's hidden state at each layer.

Training signal coupling. *Decomposed* objectives (BCE) treat each neuron independently, optimizing per-element classification against an oracle mask. *Intermediate* objectives (MSE on hidden states) capture local coupling within a single layer but not end-to-end output effects. *Coupled* objectives (KL on output distribution) propagate gradients through the full forward pass; each mask decision's gradient reflects its joint effect with all other masks on the final distribution.

SPON (Xu et al., 2025) distills a per-layer bias via KL but cannot adapt to per-input neuron importance (static × coupled). DejaVu (Liu et al., 2023) and FastForward (Gautam et al., 2026) achieve per-token granularity but use BCE, where each neuron's gradient is independent of other mask states. D2DMoE (Szatkowski et al., 2024) provides intermediate coupling (MSE on hidden states) but not end-to-end output effects. We define *coupling alignment* as whether the training loss optimizes a quantity that composes across layers. KL on output logits directly optimizes next-token prediction loss, which does not suffer from error composition; per-layer MSE minimizes reconstruction error independently at each layer, but layer-wise optima do not compose—a mask minimizing FFN output error at layer ℓ may amplify errors in layers $\ell{+}1 \ldots L$. The SNR analysis (§3.6) corroborates this: KL achieves the best masks despite 2.3× lower SNR than BCE, while MSE produces the worst masks. The framework predicts that losses operating on output-level quantities should outperform intermediate-level losses, a hypothesis consistent with our MSE results and testable with alternative output-level objectives.

P Oracle Mask Provenance

C1's oracle masks are extracted from a converged KL predictor and binarized via hard threshold (logits > 0, equivalent to $\sigma > 0.5$); they reflect one specific training trajectory (optimizer state, data order, Gumbel noise). The s42/s123/s456 consistency ($\leq$1.70pp) provides partial evidence of robustness, but does not rule out trajectory-dependent confounds.

Q SPON Orthogonality

SPON (Xu et al., 2025) bias correction is weakly correlated with the KL predictor (cosine similarity 0.196, R^2=0.038) and yields only 2.4% PPL improvement (12.08→11.79); the predictor already captures most recoverable signal. The decomposition confirms that SPON and the KL predictor operate on largely orthogonal error sources.

Table 15: Full allocation analysis: global top-k vs. uniform across sparsity levels (LLaMA-3.1-8B). Entries with $\pm$: 3-seed mean$\pm$std ($s \in \{42, 123, 456\}$); others: seed 42. *High MMLU seed variance (individual seeds span 25.3–43.5%).

Sparsity	Allocation	PPL↓	ARC-c	WG	MMLU	GSM8K
25%	Global	**8.17**	45.4	70.6	**46.6**	**11.5**
	Uniform	8.23	**46.5**	**71.8**	44.0	11.0
30%	Global	8.96±0.02	**43.8±3.6**	**70.2±0.9**	35.7±9.5*	**8.0±0.7**
	Uniform	**8.81±0.02**	43.7±2.5	68.9±0.4	**40.7±1.3**	7.5±0.1
35%	Global	**9.31**	**38.9**	**67.6**	**40.4**	**7.7**
	Uniform	9.44	38.7	66.5	37.8	4.3
40%	Global	**10.14**	39.5	**67.5**	**39.5**	**6.6**
	Uniform	10.76	**39.6**	63.1	35.5	3.6
45%	Global	**10.80**	**39.2**	**64.6**	30.4	**3.0**
	Uniform	10.95	37.6	63.6	**30.6**	2.7
50%	Global	**11.79**	**32.85**	**63.46**	**28.1±0.2**	1.29
	Uniform	11.92±0.07	32.76	62.90	28.0±0.6	**2.12**

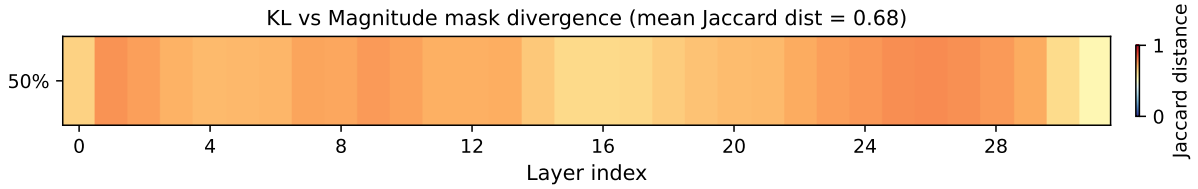


Figure 7: Per-layer Jaccard distance between KL-derived and magnitude-based masks at 30% and 50% sparsity (64 samples, 131K tokens). Divergence peaks in layers 24–28 (Jaccard distance >0.73 at 50%) and increases with sparsity across all layers.

Table 16: Cross-architecture comparison (WikiText-2 PPL↓). Adv = TEAL / KL ratio. **Note:** LLaMA results use global top-k allocation; Mistral results use uniform allocation. Mistral 30%: 3-seed mean$\pm$std; other rows: seed 42.

	Mistral-7B			LLaMA-8B		
	KL	TEAL	Adv	KL	TEAL	Adv
30%	**6.93±0.19**	8.38	1.21×	**8.96**	10.97	1.22×
50%	**8.65**	18.31	2.12×	**11.79**	23.31	1.98×
70%	**13.33**	61.72	4.63×	**17.91**	100.11	5.59×

Table 17: Mistral-7B-v0.3 downstream evaluation at 30% sparsity. $\pm$: 3-seed mean$\pm$std ($s \in \{42, 123, 456\}$); TEAL is deterministic (single run). ARC-c exhibits high seed variance (σ=6.4pp).

Method	PPL↓	ARC-c	WG	HS	PIQA	MMLU	GSM8K
Dense	5.52	60.2	78.7	83.2	81.9	62.4	36.5
KL 30%	**6.93±0.19**	**44.9±6.4**	**70.2±1.2**	**69.7±0.8**	**74.8±0.4**	43.3±1.5	8.6±2.2
TEAL 30%	8.38	41.1	63.5	60.1	72.1	**43.5**	**15.9**

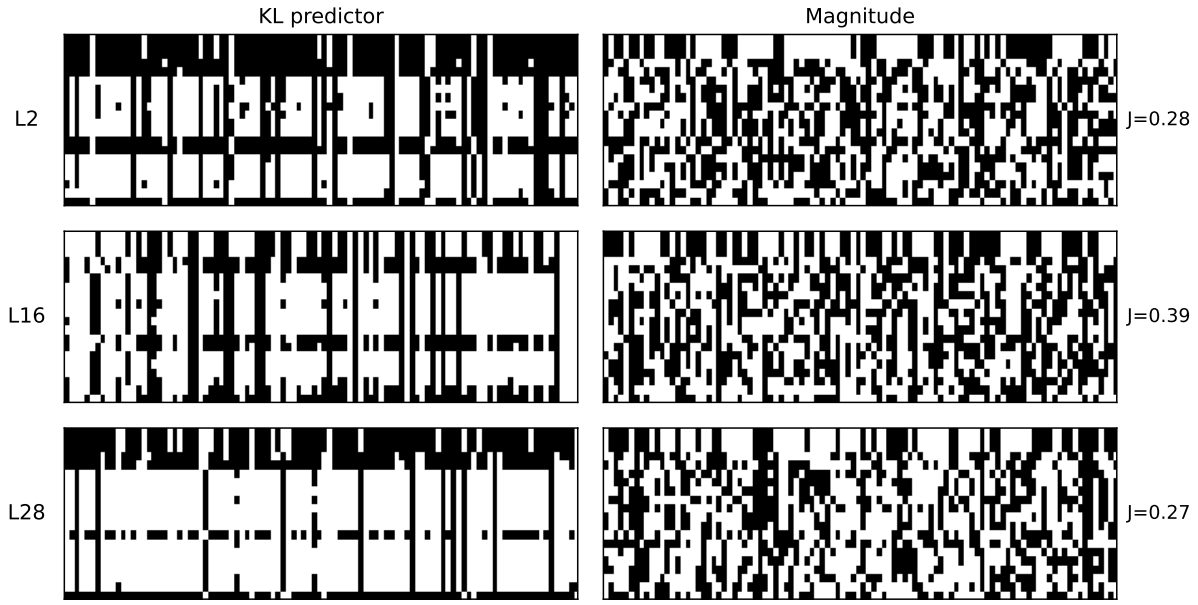


Figure 8: Binary mask comparison between KL-derived and magnitude-based masks at selected layers (L2, L16, L28) for 30% and 50% sparsity (64 samples). Black = selected by both, light regions = selected by only one method. Deep layers show the largest divergence (L28 Jaccard distance: 0.60 at 30%, 0.74 at 50%).